

OneAD: Raising online video ad value for brands with AI and machine learning

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OneAD enables brands to identify and target consumers with relevant online video advertisements leveraging the state-of-the-art infrastructure, efficient computing power, and delivery of Google AdSense to extract value from volumes of data.



About OneAD

Founded in 2010, OneAD is one of the biggest video ad operators in Taiwan. The business applies core competencies in advertising technology, media technology, and data technology to deliver data-driven advertising solutions that help clients build their brands. OneAD has delivered compelling video ad formats to premium publishers and owns large volumes of data that enable clients to profile consumers and enrich the value of their advertisements.

Google Cloud results

- Processes data collected from 99.8% of the digital audience in Taiwan
- Enables businesses to target specific customer groups with video advertisements
- Achieves near-100% uptime to meet customer requirements

Spending advertising and marketing in a fragmented digital market is a challenge for many businesses. Taipei-headquartered OneAD (https://www.onead.com.tw/) uses Google Cloud and data technologies to help clients in the automotive, and other verticals connect with consumers most likely to buy their products and services. OneAD's partners include major TV and top media websites.

OneAD media technology products include TV and mobile advertising formats,

Reduced
to 10%
moderate
hour

Industries: Other

Location: Taiwan

Platform



About iKala

iKala is a Taiwan-based Google Cloud Partner. The business has more than four years of experience and has delivered services including technical consulting, billing support, service migration, and enterprise training to over 200 businesses in more than 12 industries across APAC. iKala has more than 40 professional architects with over 20 cloud certifications.



BigQuery (<https://cloud.google.com/bigquery/>)

ABOUT IKALA

Cloud Storage (<https://cloud.google.com/storage/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

TensorFlow (<https://www.tensorflow.org/>)

services including technical consulting

AI Platform (<https://cloud.google.com/ai-platform/>)

Cloud Dataproc

(<https://cloud.google.com/dataproc/>)

iKala has more than 40 professional

(software-based) purchasing of advertising, and technologies that demand from advertisers wanting application or website to market services (allowing publishers to stimulate competition and increase revenue).

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud Video Intelligence API

(<https://cloud.google.com/video-intelligence/>)

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*—Ethan Yeh, Data
Scientist, OneAD*

OneAD also delivers artificial intelligence (AI)-powered algorithms that enable publishers to forecast media traffic and sell as many advertising positions as possible on their websites or applications. Furthermore, the business provides an open framework that enables advertisers and media agencies to use any tracking code they like to measure performance.

OneAD's advertising technology products include partnerships with advertising exchanges such as Google AdX and demand-side platforms such as Adobe and DoubleClick Bid Manager. The business provides programmatic buying services to a range of international customers, including liquor, luxury, smartphone, and banking brands.

In addition, OneAD has created a data technology service called OneID that uses machine learning (ML) algorithms to capture information about consumers' browsing behavior and online journeys.

Since it was founded in 2010, OneAD has grown rapidly. Today, the company reports that its video advertising services reach 99.8 percent of the digital audience in Taiwan.

Big data, AI, and ML

OneAD relies heavily on big data, AI, and ML to meet customer needs and differentiate itself from competitors. The business runs a data science team that uses technology to identify the attributes of consumers who interact with advertisements.

“We process data collected from our coverage of 99.8 percent of the digital audience in Taiwan every day, so we need a highly scalable and cost-effective platform. Cloud Dataproc enables us to manage our services easily and quickly while enabling us to pay only for the resources

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—Ethan Yeh, Data
Scientist, OneAD

“Our team uses technology to collate, clean, query, and analyze data to identify the age, location, and other attributes of consumers who view or click on a mobile or web advertisement,” explains Ethan Yeh, Data Scientist, OneAD. OneAD uses this data to develop the ML models and algorithms that help businesses place advertisements to target desired groups of consumers and underpin the OneID service.

Google the best fit

OneAD started operations using a cloud service based in Asia. However, after using the service for a year, the business decided to review its options. Surging growth, client demands for more precise, insightful targeting of consumers, and the need to deliver near-100 percent uptime meant OneAD needed to migrate to a scalable, highly available cloud service with deep ML and AI capabilities.

OneAD worked with Google Cloud Partner [iKala](https://gcp.expert/en/) (https://gcp.expert/en/) – whose representatives had established a relationship with the advertising and media business while it was reviewing its options – to identify the most appropriate cloud service. “We found Google Cloud was the best fit for our

operations,” explains Yeh. “Google is a globally recognized technology leader with a proven track record in ML, AI, and the cloud. The company has a high-quality infrastructure that can support high-performance advertising delivery systems.”

OneAD then worked with iKala to create a Google Cloud (<https://cloud.google.com/>) architecture that could meet the demands of the business and its clients. The partner’s engineers and account teams were well versed in the platform and helped ensure the project was delivered on time and within budget.

OneAD initially used Compute Engine (<https://cloud.google.com/compute/>) to provide infrastructure as a service for its advertising and media delivery and reporting systems. Compute Engine enabled the business to quickly create virtual machines to run these systems and take advantage of highly scalable, load-balanced compute power to manage variable demand. OneAD also opted to run Apache Spark and Hadoop clusters on Cloud Dataproc (<https://cloud.google.com/dataproc/>) to process and analyze data residing in a BigQuery (<https://cloud.google.com/bigquery/>) warehouse.

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Fast, responsive delivery

“We process data collected from our coverage of 99.8 percent of the digital audience in Taiwan every day, so we need a highly scalable and cost-effective platform,” says Yeh. “Cloud Dataproc enables us to manage our services easily and quickly while enabling us to pay only for the resources we use – down to the second.” All files and information then reside in a highly secure environment in Cloud Storage (<https://cloud.google.com/storage/>).

OneAD uses the data it captures to inform its ML models and algorithms. The business has deployed TensorFlow (<https://www.tensorflow.org/>), the open source software library originally developed by the Google AI organization, to operate as the foundation of its ML architecture. It then chose to use Cloud Machine Learning Engine (<https://cloud.google.com/ml-engine/>) to train its models in TensorFlow at scale without having to manage any infrastructure – improving its

performance in helping customers predict consumer demographics.

Machine Learning training times reduced

“Moving to Cloud Machine Learning Engine enabled us to reduce the time required to train our OneID ML models from three hours to just one hour,” says Yeh. The business is also delivering the uptime needed to meet customers’ demands for timely insights into consumer behavior and demographics.

With Google Cloud services established and delivering value to the business, OneAD is considering adopting a [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine/) (https://cloud.google.com/kubernetes-engine/) containerized application environment to reduce costs and [Cloud Video Intelligence](https://cloud.google.com/video-intelligence/) (https://cloud.google.com/video-intelligence/) to help identify the characteristics of a video advertisement that will resonate with targeted consumers. Cloud Video Intelligence enables businesses to search and discover video content by extracting metadata through an API.

The business has monthly meetings with Google and iKala to discuss Google Cloud developments and identify new opportunities. “Just recently, we worked with Google to evaluate the potential of APIs to deliver further benefits to our business,” says Yeh. “In addition, our account manager at iKala, is always available to address any issues or answer any questions for us.”

